Network Inference from Deterministic Dynamics: From Heat Diffusion to SI Epidemics and Partial Observations

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2022102070
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***Abstract*—Inferring network structure from time-varying processes is a central inverse problem at the intersection of physics, signal processing, and epidemiology. Foundational work from nonlinear dynamics has shown that interaction topology may be recovered from deterministic system responses. Graph signal processing has further demonstrated that operator-based models, such as the heat equation on graphs, yield linear inverse problems in the unknown Laplacian. Building on these ideas, this report reproduces a recent heat-diffusion network learning method and extends the deterministic framework to the nonlinear Susceptible–Infected (SI) epidemic model. The primary contribution of this work is a conceptual and methodological extension to partial observations, enabling network inference when only a subset of node trajectories is observable. Experimental results are provided for diffusion-based learning and full-observation SI dynamics, while the partial-observation methodology is introduced and reserved for detailed algorithmic development.**

I. Introduction

Network inference seeks to recover the underlying connectivity of a system from observations of its temporal evolution. When the processes governing each node's evolution are deterministic, the interaction structure becomes encoded in the observed trajectories and their derivatives. This viewpoint is formalized in general dynamical-systems–based network reconstruction frameworks [1], where node dynamics of the form

$$\dot{x}_i = f_i(x_i) + \sum_j J_{ij} g_{ij}(x_i, x_j) \tag{1}$$

implicitly contain information about the unknown interactions J_{ij}. Under suitable variation in state trajectories, these relationships can be assembled into solvable inverse problems.

Within the graph-signal–processing community, Nannuru *et al.* [2] demonstrated that this principle becomes particularly tractable in the setting of diffusion processes on graphs. Diffusion dynamics are linear in the graph operator, governed by the heat equation $\dot{\mathbf{x}} = -L\mathbf{x}$, enabling the underlying Laplacian to be learned via linear least-squares methods. Their formulation uses temporal derivative approximations and half-vectorization to transform the Laplacian learning task into a structured linear inverse problem. Because diffusion is completely linear in both state and operator, this provides a clean baseline model for studying deterministic network inference.

However, many real-world networks do not evolve according to linear diffusion. Epidemic spreading, information contagion, and influence propagation processes typically follow nonlinear dynamics. A widely studied example is the Susceptible–Infected (SI) model [3], where node states represent infection probabilities and edges represent interaction or contact strengths. SI dynamics are nonlinear in state but *linear in adjacency*, suggesting that with suitable reformulation, linear regression methods can again be applied for network inference. Unlike diffusion, SI processes are not governed by closed-form semigroup operators: state trajectories must be simulated numerically, and the regression structure must be extracted directly from the dynamics.

An additional complication arises in realistic settings: one may not be able to observe all nodes. Partial observation introduces missing states and derivatives into the regression, resulting in underdetermined systems requiring structured constraints or redesign of the inference method.

This report addresses these considerations by developing a unified workflow for deterministic network inference under diffusion and SI dynamics, including both full- and partial-observation settings. The contributions are as follows:

1) We reproduce the diffusion-based graph-learning method of Nannuru *et al.* [2], following their sampling strategy, numerical-derivative approximation, and half-vectorized least-squares formulation.
2) We design a new inference algorithm for SI dynamics under full observation, using the continuous model in [3] and the linear regression structure arising from the multiplicative term $(1 - x_i)x_j$. Our formulation uses vectorization and Kronecker identities (detailed in the methodology section) but avoids unnecessary tensor expansions.
3) We extend the inference problem to *partial observation*, formalizing what it means to observe only a subset of node trajectories and characterizing the form of the resulting partially observable SI equations. This leads

naturally to new identifiability constraints and a modified regression pipeline.

The methodology section describes each component of the workflow: the deterministic network-inference setting, the diffusion-based method, the SI-based extension, and the partial-observation formulation.

II. Background

A. Deterministic Network-Inference Setting

Network dynamics over N nodes are described generally by

$$\dot{x}_i(t) = f_i(x_i(t)) + \sum_j A_{ij}\, g_{ij}(x_i(t), x_j(t)), \tag{2}$$

where A is the unknown adjacency (or Laplacian-derived) matrix to be inferred. As discussed in [1], when trajectories $(x_i(t), \dot{x}_i(t))$ are observed, Eq. (2) induces constraints on A. If the right-hand side is linear (or linearizable) in A, the inference reduces to a structured inverse problem.

The diffusion and SI processes examined in this report both fit into this framework but differ substantially in linearity and identifiability.

B. Scalar Diffusion Dynamics

The classical heat equation describes scalar diffusion:

$$\frac{\partial u}{\partial t} = \kappa \Delta u, \tag{3}$$

with u spreading smoothly across space. This is a linear PDE whose solution is obtained through the heat kernel.

C. Diffusion on Graphs

On a graph, spatial derivatives are replaced by the graph Laplacian $L = D - A$, giving

$$\dot{\mathbf{x}}(t) = -L\, \mathbf{x}(t). \tag{4}$$

The solution has closed form:

$$\mathbf{x}(t) = e^{-Lt}\, \mathbf{x}(0), \tag{5}$$

and the temporal derivative satisfies a linear equation in L, which is the basis of diffusion-based network inference. Because the operator enters linearly, derivative measurements provide linear equations in the unknown Laplacian entries.

D. Scalar SI Dynamics

The scalar SI model describes infection probability $x(t)$ evolving as:

$$\dot{x}(t) = \beta x(t)\left(1 - x(t)\right), \tag{6}$$

a nonlinear logistic-type growth. The nonlinearity corresponds to susceptible–infected interaction within a well-mixed population.

E. SI Dynamics on Graphs

To incorporate heterogeneous contact patterns, SI spreads over a network via

$$\dot{x}_i(t) = \beta\left(1 - x_i(t)\right) \sum_j A_{ij} x_j(t). \tag{7}$$

Unlike diffusion, SI is nonlinear in state but *affine in adjacency*. Infection spreads directionally along edges, so A rather than L is the natural operator of interest. Because $(1 - x_i)x_j$ evolves in time, SI produces a time-varying linear-in-parameters system that motivates the LS2 framework described in the methodology.

F. Partial Observation Setting

Suppose only nodes $\mathcal{O} \subseteq \{1, \ldots, N\}$ are observed. For $i \in \mathcal{O}$,

$$\dot{x}_i(t) = \beta(1 - x_i(t))\left(\sum_{j\in\mathcal{O}} A_{ij}x_j(t) + \sum_{j\notin\mathcal{O}} A_{ij}x_j(t)\right). \tag{8}$$

The second summation contains unobserved states and produces an underdetermined system whose structure must be exploited to recover the adjacency. The remainder of the report develops the algorithmic strategy and identifiability considerations for this partial-observation regime.

III. Methodology

A. Diffusion Dynamics and Laplacian Learning (Reproduction of [2])

1) Dynamics Generation and Sampling: Graph diffusion obeys

$$\dot{\mathbf{x}}(t) = -L\mathbf{x}(t). \tag{9}$$

Since the solution is closed form,

$$\mathbf{x}(t) = e^{-Lt}\mathbf{x}(0), \tag{10}$$

we may generate trajectories at arbitrarily high resolution and sample them uniformly in time. Following [2], we approximate derivatives via

$$\mathbf{v}_i = \frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{\Delta t}, \qquad \mathbf{u}_i = -\mathbf{x}_i. \tag{11}$$

Each sample produces

$$L\mathbf{u}_i = \mathbf{v}_i. \tag{12}$$

2) Vectorization and Half-Vectorization: Writing L in vech-form produces

$$U_i\, \ell = \mathbf{v}_i, \qquad \ell = \mathrm{vech}(L). \tag{13}$$

Stacking in time yields

$$U\, \ell = \mathbf{V}, \tag{14}$$

a least-squares problem with Laplacian constraints. Half-vectorization reduces unknowns from N^2 to $N(N+1)/2$, enforces symmetry, improves conditioning, and matches the true degrees of freedom.

B. SI Dynamics and Symmetric LS (LS2) Framework

1) Dynamics and Linear Structure: SI evolution over graphs is governed by Eq. (7). For each node,

$$\dot{x}_i(t) = \beta(1 - x_i(t))\, \mathbf{x}(t)^\top \mathbf{a}_i, \tag{15}$$

where $\mathbf{a}_i$ denotes row i of the adjacency. Although nonlinear in state, this is affine in $\mathbf{a}_i$. A regression-like formulation is therefore possible, but solving rows independently fails to enforce global symmetry ($A_{ij} = A_{ji}$). LS2 addresses this.

2) Global Vectorized SI Equation: Using the identity

$$A\mathbf{x}_t = (I \otimes \mathbf{x}_t^\top)\, \mathrm{vec}(A), \tag{16}$$

the SI system becomes

$$\dot{\mathbf{x}}_t = \beta(I - \mathrm{diag}(\mathbf{x}_t))(I \otimes \mathbf{x}_t^\top)\mathrm{vec}(A). \tag{17}$$

3) Symmetry via Duplication Matrix: For undirected networks, A is symmetric. Using the duplication matrix D_N:

$$\mathrm{vec}(A) = D_N\, \mathrm{vech}(A), \tag{18}$$

reduces the parameter space to $N(N+1)/2$ variables and imposes symmetry exactly. Substituting this into (17) gives the reduced operator

$$\tilde{F}_t = (I - \mathrm{diag}(\mathbf{x}_t))(I \otimes \mathbf{x}_t^\top)D_N, \tag{19}$$

leading to

$$\frac{\Delta \mathbf{x}_t}{\beta} \approx \tilde{F}_t\, \mathrm{vech}(A). \tag{20}$$

4) LS2 Estimation System: Stack all time points (and all processes):

$$\mathbf{v} = \beta \tilde{F}\, \mathrm{vech}(A), \tag{21}$$

and solve using least squares:

$$\widehat{\mathrm{vech}(A)} = \tilde{F}^+ \left(\frac{\mathbf{v}}{\beta_{\text{init}}} \right). \tag{22}$$

Reconstruct:

$$\hat{A} = \mathrm{vech}^{-1}(\widehat{\mathrm{vech}(A)}), \tag{23}$$

with post-processing ensuring $A_{ii} = 0$, non-negativity, and exact symmetry.

LS2 is therefore the natural SI analogue of the half-vectorized diffusion learner: both exploit linear-in-operator dynamics, but LS2 accounts for time-varying interactions via $(1 - x_i(t))$ and enforces global structure through duplication-matrix–based reduction.

C. Learning SI Networks Under Partial Observability

We now consider the setting in which only a subset of node states $\mathcal{O} \subseteq \{1, \ldots, N\}$ is observed, with $N_{\text{obs}} = |\mathcal{O}|$. Partition the adjacency matrix as

$$A = \begin{bmatrix} A_{OO} & A_{OH} \\ A_{HO} & A_{HH} \end{bmatrix},$$

where $A_{OO} \in \mathbb{R}^{N_{\text{obs}} \times N_{\text{obs}}}$ and $A_{OH} \in \mathbb{R}^{N_{\text{obs}} \times N_{\text{hid}}}$ influence observable SI dynamics.

For each observed node $i \in \mathcal{O}$, the continuous SI model becomes

$$\dot{x}_i(t) = \beta(1 - x_i(t)) \left(\sum_{j \in \mathcal{O}} A_{ij} x_j(t) + \sum_{j \in H} A_{ij} x_j(t) \right), \tag{24}$$

where the second summation involves both unknown adjacency weights and unobserved states, and therefore must be modeled indirectly.

1) Step 0: Initialization: The learning algorithm requires initial estimates of A_{OO}, W, and Z. We employ the following procedure.

(a) LS2-based initialization of A_{OO}: Using midpoint states

$$x_{i,t}^{\text{mid}} = \frac{1}{2}(x_{i,t} + x_{i,t+1}), \qquad \dot{x}_{i,t} \approx \frac{x_{i,t+1} - x_{i,t}}{\Delta t},$$

define the observable linear quantity

$$Y_{i,t} = \frac{\dot{x}_{i,t}}{\beta\big(1 - x_{i,t}^{\text{mid}}\big)}.$$

Under SI dynamics this satisfies

$$Y_{i,t} = \sum_{j \in \mathcal{O}} A_{ij} x_{j,t}^{\text{mid}} + \sum_{j \in H} A_{ij} x_{j,t}^{\text{mid}}.$$

Ignoring the hidden contribution yields an LS2-style half-vectorized least-squares approximation

$$U\, \mathrm{vech}(A_{OO}) \approx Y,$$

where

$$U_t = (I \otimes (x_t^{\text{mid}})^\top)D,$$

and D is the duplication matrix ensuring symmetry. After solving, we project:

$$A_{OO} \leftarrow \max(0, A_{OO}), \qquad A_{OO} \leftarrow A_{OO} - \mathrm{diag}(\mathrm{diag}(A_{OO})).$$

(b) NMF-based initialization of W and Z: Form residuals

$$R_{:,s} = \left| Y_t - A_{OO}\, x_t^{\text{mid}} \right|, \qquad s = (t, \text{process}),$$

stacked across all SI processes. Apply nonnegative matrix factorization:

$$R \approx W_0 Z_0,$$

and normalize columns of W_0, rescaling Z_0 accordingly.

(c) Random initialization variant: For robustness analysis, we also test

$$A_{OO}^{(0)} \sim \mathrm{Uniform}(0, 1)$$

. W and z are initialized similar to the previous section.

2) Step 1: Observable Linear Measurements: Using midpoint states, define

$$Y_{i,t} = \frac{\dot{x}_{i,t}}{\beta\big(1 - x_{i,t}^{\text{mid}}\big)} = \sum_{j \in \mathcal{O}} A_{ij} x_{j,t}^{\text{mid}} + \sum_{j \in H} A_{ij} x_{j,t}^{\text{mid}}. \tag{25}$$

Stacking over observed nodes gives

$$\mathbf{Y}_t = A_{OO} \mathbf{x}_t^{\text{mid}} + A_{OH} \mathbf{x}_{H,t}^{\text{mid}}. \tag{26}$$

Since $\mathbf{x}_{H,t}^{\text{mid}}$ is unobserved, the second term must be modeled implicitly.

3) Step 2: Low-Rank Latent Representation of Hidden Influence: We approximate the hidden contribution by a compact low-rank factorization:

$$A_{OH}\mathbf{x}_{H,t}^{\mathrm{mid}} \approx W\,\mathbf{z}_t, \tag{27}$$

where

- $W \in \mathbb{R}^{N_{\mathrm{obs}}\times K}$ represents how each observed node responds to K latent sources,
- $\mathbf{z}_t \in \mathbb{R}^K$ is the latent influence at time t,
- K is small (typically 1–3 here).

Across multiple SI trajectories with different initial conditions, the hidden-node contribution exhibits low-rank structure, motivating this representation.

4) Step 3: Latent-State Estimation (Z-update): With A_{OO} and W fixed, (26) and (27) give

$$\mathbf{Y}_t \approx A_{OO}\mathbf{x}_t^{\mathrm{mid}} + W\mathbf{z}_t.$$

Thus,

$$\mathbf{z}_t = \arg\min_{\mathbf{z}} \left\|\mathbf{Y}_t - A_{OO}\mathbf{x}_t^{\mathrm{mid}} - W\mathbf{z}\right\|_2^2. \tag{28}$$

Since K is small, we obtain the closed-form update

$$\mathbf{z}_t = (W^\top W)^{-1}W^\top\left(\mathbf{Y}_t - A_{OO}\mathbf{x}_t^{\mathrm{mid}}\right). \tag{29}$$

5) Step 4: Joint Adjacency–Latent Update (A–W update): After updating all latent variables $\{\mathbf{z}_t\}$, we refine both A_{OO} and W simultaneously. Stacking all processes and all time samples yields the global linear model

$$\mathbf{Y} = \Phi_A\,\mathrm{vech}(A_{OO}) + \Phi_W\,\mathrm{vec}(W), \tag{30}$$

where the operators are:

$$\Phi_A = (I \otimes (x_t^{\mathrm{mid}})^\top)D,$$
$$\Phi_W = (Z^\top \otimes I).$$

Unlike the symbolic LS2 operator appearing in diffusion and in the general SI formulation, the factor $(I - \mathrm{diag}(x_t^{\mathrm{mid}}))$ is **not included** here because it is already absorbed into the definition of Y_t in (25). Thus, (25) is exactly linear in the adjacency rows and requires no additional diagonal correction.

We solve the regularized least-squares system

$$\theta^{\mathrm{new}} = (\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top\mathbf{Y}, \tag{31}$$

where θ concatenates $\mathrm{vech}(A_{OO})$ and $\mathrm{vec}(W)$, and λ is a small ridge penalty ensuring numerical stability.

After reshaping,

$$A_{OO}^{\mathrm{new}} = \mathrm{vech}^{-1}(\theta_A), \qquad W^{\mathrm{new}} = \mathrm{mat}(\theta_W).$$

We apply structural projections:

$$A_{OO} \leftarrow \max(0, A_{OO})$$
$$A_{OO} \leftarrow A_{OO} - \mathrm{diag}(\mathrm{diag}(A_{OO}))$$
$$W \leftarrow \max(0, W).$$

6) Step 5: Alternation and Convergence: The algorithm alternates:

$$\boxed{\text{Z-update} \quad \longleftrightarrow \quad \text{A–W update}}$$

Convergence is monitored by

$$\|\mathbf{Y} - \Phi\theta\|_2^2, \qquad \frac{\|A_{OO}^{\mathrm{new}} - A_{OO}\|_F}{\|A_{OO}\|_F}, \qquad \frac{\|W^{\mathrm{new}} - W\|_F}{\|W\|_F}.$$

Multiple SI processes with different initial conditions are essential for effective rank in the global design matrix and stable recovery.

7) Interpretation: The method yields a structured decomposition of the observable SI dynamics:

- A_{OO} reconstructs the portion of the spreading structure attributable to observed nodes,
- WZ captures the influence transmitted through unobserved nodes in a compact low-rank form,
- alternating Z-updates and A–W updates progressively distributes reconstruction responsibility between explicit edges and latent factors.

Thus, the proposed partial-observation reconstruction extends LS2 to incomplete data while preserving symmetry-enforcing half-vectorization, global least-squares inference, and deterministic SI modeling.

D. Role of Multiple Processes.

All three dynamical settings considered in this report—diffusion, fully observed SI, and partially observed SI—benefit strongly from using multiple independent processes. Although the equations differ, the underlying reason is the same: each system eventually stabilizes (diffusion smooths out, SI saturates to $x \to 1$), causing late-time samples to contain very little structural information. Early-time behaviour is informative, but a single trajectory provides only one "direction" of variation in the regressor (whether $\mathbf{x}(t)$ for diffusion or $(1 - x_i)x_j$ for SI). By running multiple processes with different initial conditions, we effectively oversolve the inverse problem: the combined design matrix gains rank, symmetry constraints are better enforced, and the optimizer avoids degenerate or low-information solutions. In practice, a modest number of trajectories (4–6 for diffusion, 8–12 for SI) is sufficient to ensure that the network operator and any latent factors can be recovered stably.

IV. Experimental Results

This section presents empirical validation of the proposed methods for diffusion-based Laplacian learning and SI adjacency learning under full observation. We evaluate scaling with respect to graph size, noise, sampling resolution, and number of independent processes. All experiments were implemented in Python or MATLAB; see the accompanying repository for full scripts and reproducibility details.

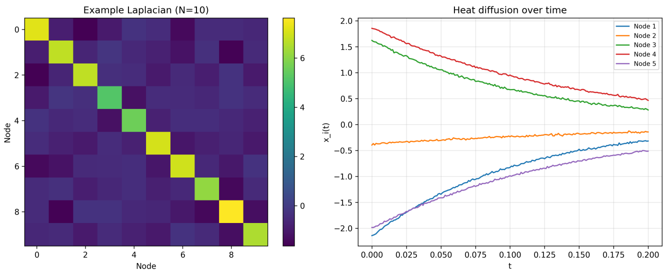

Fig. 1. Example Laplacian and Heat Diffusion process example.

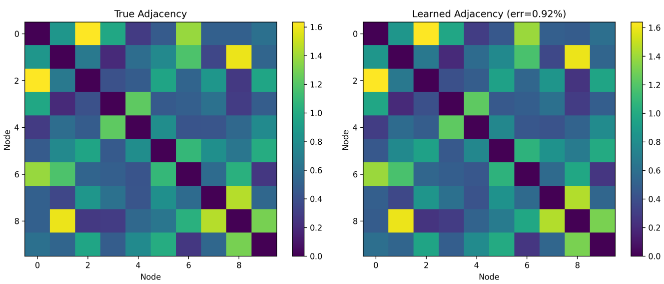

Fig. 2. Learnt Laplacian Matrix for diffusion process using the LS2 Method.

A. Diffusion-Based Learning Results

We first reproduce the canonical diffusion-network inference results of [2]. Fig. 1 shows an example ground-truth Laplacian and a heat diffusion realization generated by $\mathbf{x}(t) = e^{-Lt}\mathbf{x}(0)$.

Applying LS2 yields the reconstructed Laplacian in Fig. 2, which matches the ground truth with very low relative error. This confirms the correctness of our reproduction of the half-vectorized symmetric regression technique.

We next evaluate scaling with graph size. Fig. 3 shows that inference error increases only mildly as the number of nodes grows, indicating that LS2 remains numerically well-conditioned and benefits from the reduced number of free parameters ($N(N+1)/2$ instead of N^2).

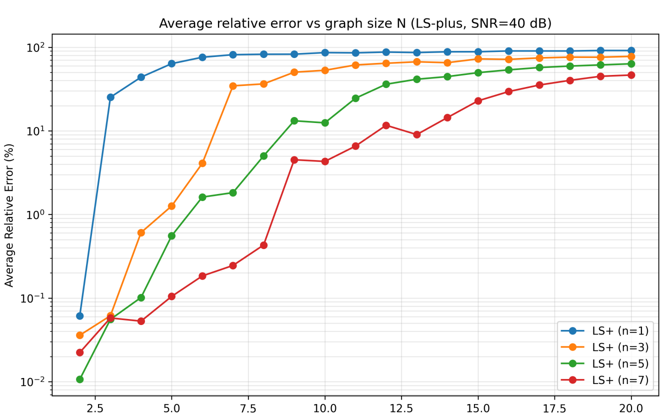

Fig. 3. Error comparison as number of nodes increases: Laplacian network inference in heat diffusion.

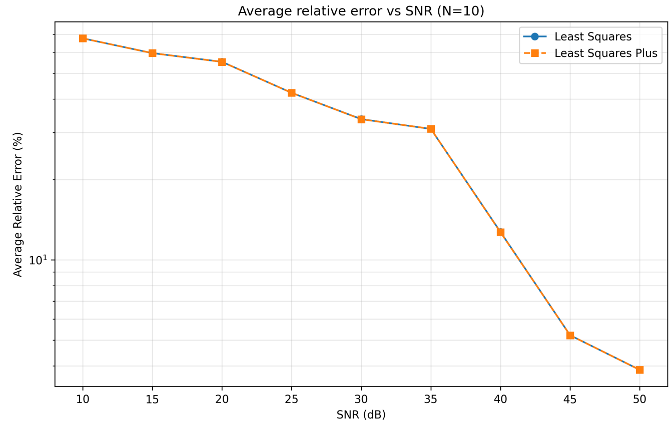

Fig. 4. Laplacian inference error (heat diffusion) when noise is added (SNR vs. error).

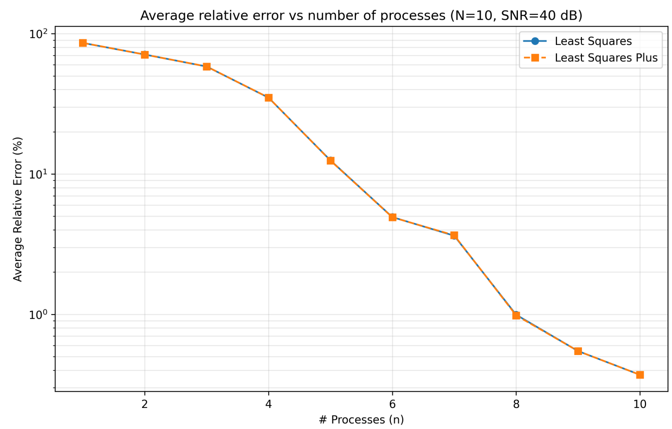

Fig. 5. Laplacian inference error (heat diffusion) as the number of simulated processes increases.

Noise robustness is shown in Fig. 4. As SNR decreases, reconstruction error increases, although LS2 remains surprisingly stable across a broad SNR range. This is consistent with the averaging effect of global least squares.

We also study the effect of the number of diffusion processes. Fig. 5 shows that increasing the number of independent initial conditions improves rank conditions and reduces inference error.

Finally, Fig. 6 illustrates the effect of sampling resolution. Finer sampling results in more accurate derivative approximations, leading to significantly improved reconstruction fidelity.

Together, these experiments validate our reproduction of the diffusion inference framework and establish LS2 as a robust and scalable estimator.

B. SI Full-Observation Results

We now present results for SI-based adjacency learning under full observability. Fig. 7 shows example SI trajectories for two independent processes on a 10-node network. All nodes eventually saturate at $x_i(t) \to 1$, but the early-time transient regime carries most of the structural information.

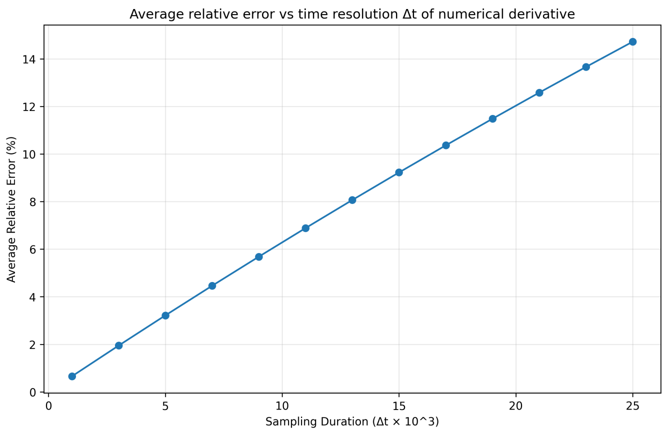

Fig. 6. Laplacian inference error (heat diffusion) as time resolution changes.

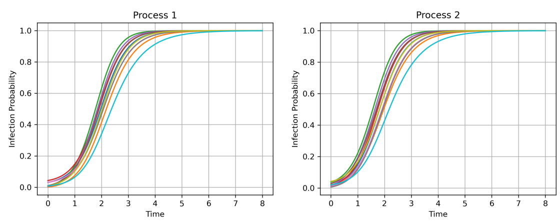

Fig. 7. Two example SI processes (10 nodes). All trajectories converge to 1 but start from small random initial infections.

A representative adjacency reconstruction is shown in Fig. 8, comparing our LS2 method with two baselines: MT (per-row regression) and LS (unconstrained least squares). In the noiseless setting, LS2 achieves the lowest error (0.26%), followed by MT (0.64%) and LS (1.01%). The benefit of half-vectorization and symmetry enforcement is evident.

Scaling with graph size is explored in Fig. 9. As N increases, LS2 maintains significantly lower error than LS and MT. This highlights the importance of structural constraints in nonlinear SI dynamics.

In Fig. 10, we evaluate which parts of the SI trajectory are most informative. Using only portions of the time axis, we find that early-time data (0–2 seconds) provide the lowest errors. Later-time windows (6–8 seconds) yield significantly worse performance, as dynamics approach saturation and offer little structural variation.

Finally, Fig. 11 compares performance under noiseless and noisy observations (40 dB SNR). Noise substantially degrades all methods, but LS2 consistently outperforms MT and LS.

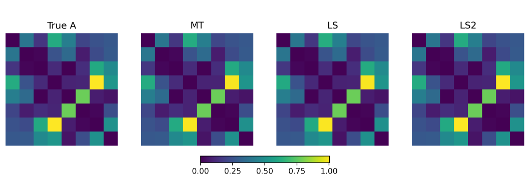

Fig. 8. Adjacency matrix inference from noiseless SI processes ($N = 8$, 20 processes). Errors: MT 0.64%, LS 1.01%, LS2 0.26%.

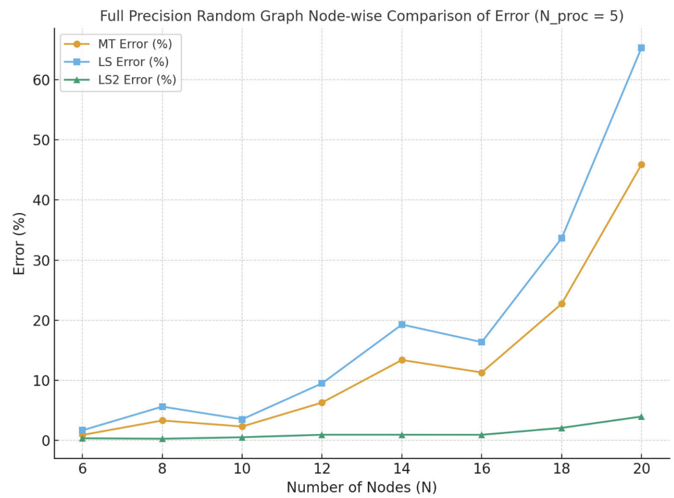

Fig. 9. SI dynamics: error scaling as number of nodes increases.

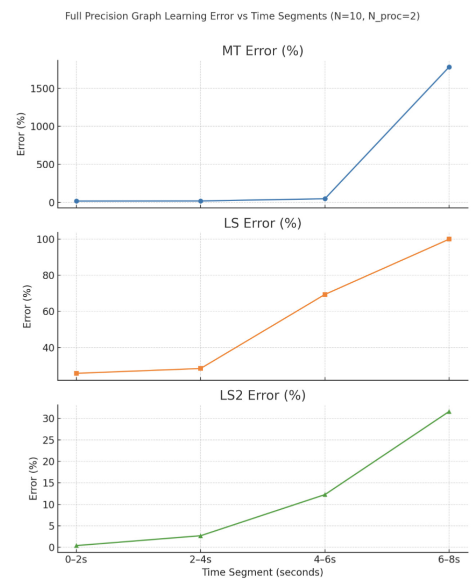

Fig. 10. Network inference error when using only selected time windows of the SI process. Early stages (0–2 s) contain the most information.

The structural constraints of LS2 act as an implicit regularizer, enhancing robustness.

Overall, the full-observation SI experiments confirm that:

- LS2 is the most accurate estimator among the tested methods,
- early-time SI dynamics are essential for identifiability,
- more SI processes significantly improve reconstruction accuracy,
- structural symmetry constraints are crucial for robustness in nonlinear dynamics.

C. Partial-Observation Results

We evaluate the proposed algorithm under partial observability with $N_{\text{obs}} = 5$ observed nodes, $N_{\text{hid}} = 2$ hidden nodes,

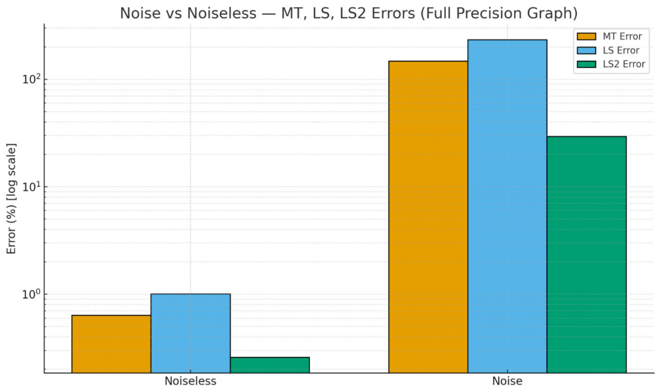

Fig. 11. SI dynamics: network inference under noiseless vs. noisy (40 dB SNR) trajectories. LS2 consistently yields lowest error.

latent rank $K = 2$, infection rate $\beta = 1$, time step $\Delta t = 0.01$, time horizon $T_{\text{end}} = 5$, and $N_{\text{proc}} = 10$ independent SI processes. The only difference between the two experiments is the initialization of A_{OO}: (i) an LS-based surrogate estimate, and (ii) a uniform random initialization. The latent factor W and the initial Z_t are identical across the two setups.

a) Why monitor objective, parameter changes, and residuals.: Because partial observability produces an underdetermined inverse problem, a single final A_{OO} is insufficient to judge whether the algorithm converged to a meaningful dynamical solution. Hence we monitor:

- **Objective descent** to check whether the model matches the SI curves.
- **A- and W-error trajectories** to see if parameter estimates move toward ground truth.
- **Z-update magnitudes** to detect latent instability or drift.
- **Residual norm** to diagnose underfitting or misspecification.

These curves jointly reveal whether the optimizer is converging toward a physically interpretable explanation of the observed SI dynamics.

b) LS-init vs. Random-init: summary of outcomes.: The LS-based initialization places the solver near the correct structural basin, producing:

- significantly faster convergence (89 iterations vs. 200),
- a lower final objective value,
- substantially lower A_{OO} and W reconstruction errors,
- smooth, monotone latent-state evolution in Z.

Random initialization begins far from the correct region of parameter space, resulting in:

- slow and noisy descent of the objective,
- unstable and large W updates,
- a much larger final W error (due to scale drift),
- higher A_{OO} error despite capturing coarse structure.

c) Objective and convergence behavior.: Both experiments reach a low objective value, showing that the SI dynamics can be fitted. However:

- **LS-init** exhibits a smooth order-of-magnitude drop followed by early saturation.
- **Random-init** shows slower descent with oscillations, indicating poorer conditioning in the A–W updates.

Thus, good initialization improves not only accuracy but also the numerical stability of the alternation scheme.

d) Quantitative comparison.: Table 1, shows the quantitative comparison of the two different methods, at convergence.

Metric	LS-init	Random-init
Iterations to converge	89	200 (maxed)
Final objective	2.66×10^{-3}	3.82×10^{-3}
A_{OO} error (%)	**8.82**	15.69
W error (%)	**36.75**	127.20
Final dA	0.088	0.157
Final dW	0.368	1.272
Final dZ	2.9×10^{-3}	5.4×10^{-3}

TABLE I
COMPARISON OF LS-INITIALIZED VS. RANDOM-INITIALIZED PARTIAL-OBSERVATION EXPERIMENTS. LS INITIALIZATION YIELDS MORE ACCURATE RECOVERY AND MORE STABLE CONVERGENCE.

e) Interpreting the WZ ambiguity.: Because the latent representation satisfies

$$WZ = (cW)(Z/c),$$

W and Z are individually unidentifiable up to a scaling factor. LS initialization implicitly selects a consistent convention, while random initialization leads to scale drift, producing high W error even when the product WZ fits the data accurately.

f) Scaling with the number of observed nodes.: We also vary N_{obs} while keeping $N_{\text{hid}} = 2$. For $4 \leq N_{\text{obs}} \leq 6$, the A_{OO} error remains low, indicating that the latent rank $K = 2$ is expressive enough to absorb hidden-node influence. However, when N_{obs} increases further (e.g., $N_{\text{obs}} = 8$), the error rises sharply. A likely explanation is that the fixed latent rank cannot model the richer set of hidden-to-observed interactions, causing part of the hidden influence to be misattributed to A_{OO} ("latent leakage"). This suggests that K should scale with N_{obs} for reliable recovery in larger observed partitions.

V. DIFFICULTIES FACED

(1) Incorrect latent factorization dimensions

In the first implementation of the partial–observability model, the latent factorization was constructed with the wrong matrix shapes. The intended structure was $W \in \mathbb{R}^{N_{\text{obs}} \times K}$ and $Z_t \in \mathbb{R}^K$, but the code mixed observed and hidden dimensions. As a result, WZ_t did not represent the hidden-node contribution correctly, and the optimizer compensated with unstable scaling in both W and Z.

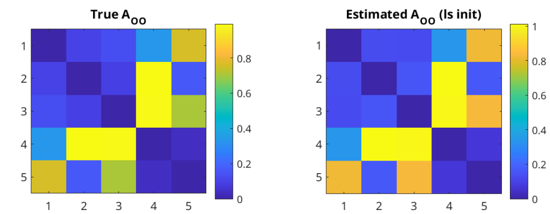

Fig. 12. **Recovered A_{OO} under LS initialization.** Edge weights closely match the ground truth, with preserved symmetry and correct magnitude ordering.

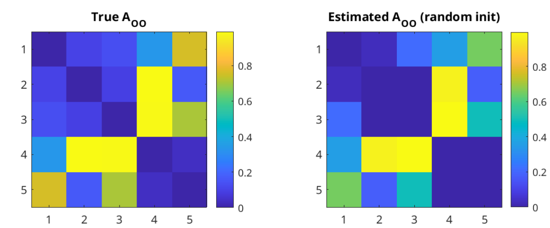

Fig. 13. **Recovered A_{OO} under random initialization.** Coarse structure is captured but many edge magnitudes collapse or distort, reflecting poor initialization.

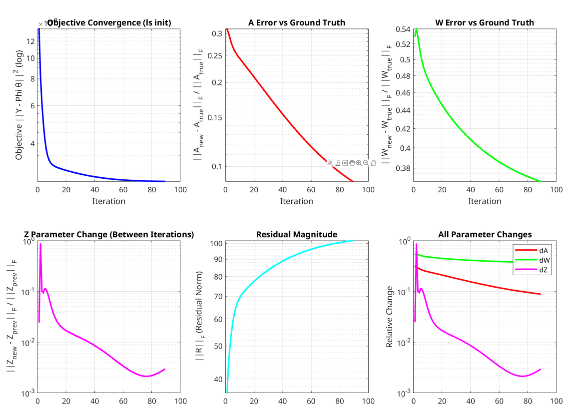

Fig. 14. **Convergence curves (LS initialization).** Smooth objective descent, shrinking parameter updates, and steady improvement in reconstruction error.

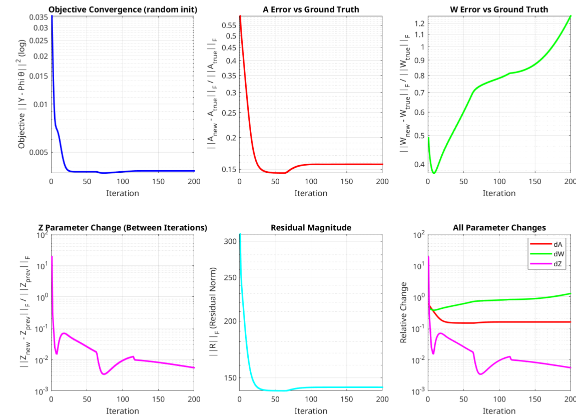

Fig. 15. **Convergence curves (Random initialization).** Objective decreases but with noisier behaviour in W and Z, indicating unstable latent-factor updates.

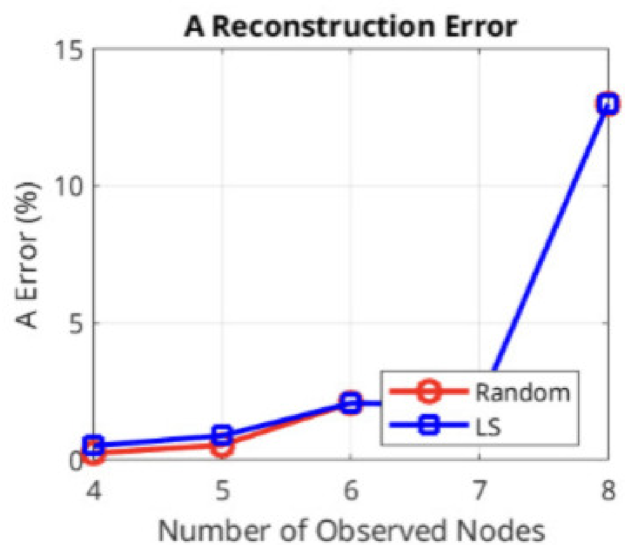

Fig. 16. **Scaling of A_{OO} error with number of observed nodes.** Errors remain low for $N_{\text{obs}} \leq 6$, but rise sharply when the latent rank becomes insufficient to model hidden-node influence.

(2) Attempting to solve for A_{OO} without vectorization

We tried to solve for A_{OO} using least squares, without vectorizing it first. This is incorrect because least squares is designed to fit parameters in a vector format rather than a matrix format. This gave us incorrect results.

VI. Future Steps

A central limitation in the current partial-observation framework is the scaling ambiguity in the latent representation:

$$WZ = (cW)(Z/c),$$

which prevents unique identification of the individual factors W and Z. Although the product WZ is sufficient to explain the hidden-node influence, recovering *interpretable* latent pathways requires breaking this ambiguity. Several directions may help address this:

- **Norm Constraints on W or Z.** Enforcing $\|W_{:,k}\|_2 = 1$ or fixing the variance of each $z_{k,t}$ would anchor the scaling and remove the free parameter c. This turns the update into a constrained least-squares problem.
- **Regularization-Based Normalization.** Adding penalties such as
$$\lambda_W \|W\|_F^2 + \lambda_Z \|Z\|_F^2$$
encourages balanced magnitudes and prevents scale drift. Jointly tuning (λ_W, λ_Z) may naturally push the optimizer toward a stable scale.
- **Temporal Smoothness Priors on Z_t.** Since hidden-node influence often changes smoothly over time, adding a penalty like
$$\gamma \sum_t \|Z_{t+1} - Z_t\|_2^2$$
restricts the degrees of freedom in Z and indirectly stabilizes the scaling of W.
- We can look into enforcing SI dynamics specific constraints.

Overall, future work will focus on incorporating these normalization or regularization strategies into the alternating updates, allowing the latent factors to become not only numerically stable but also uniquely interpretable.

VII. Conclusion

This work examined deterministic network–inference across several dynamical settings, starting from linear diffusion, moving to nonlinear SI dynamics, and finally to the more difficult partially–observed SI case. Across these systems we used the same overall idea: construct a global least–squares constraint (via LS2/half–vectorization) and test how far such deterministic structure alone can take us in recovering the underlying network.

The diffusion experiments served as a baseline where the linear model matches the LS2 assumptions exactly. As expected, recovery was reliable and well-conditioned, confirming that the formulation behaves as intended when the dynamics are linear. The full-observation SI experiments then tested the same framework for nonlinear dynamics. Even though SI eventually saturates, we found that early-time transients combined across multiple trajectories still provide enough variation for accurate recovery, showing that LS2-style constraints remain effective well outside the linear regime.

The main focus was the partially-observed SI setting, where part of the network is hidden and must be captured through a latent factorization. Here the experiments showed that the algorithm can recover meaningful structure from purely deterministic SI trajectories, but only under the right conditions. Good initialization of the observed adjacency (via LS2) was consistently important: it improved conditioning, stabilized the latent updates, and reduced both A_{OO} and W reconstruction error. Random initialization, in contrast, led to slower or unstable convergence and large errors in the latent pathway, despite still fitting the SI curves.

We also observed that when the number of observed nodes becomes large relative to the latent rank, the model begins to misattribute hidden-node influence to A_{OO}—a phenomenon we referred to as “latent leakage.” This suggests that partial-observation inference may require scaling the latent dimension with graph size or using additional constraints to separate hidden and observed pathways.

Overall, the experiments indicate that deterministic SI dynamics do encode recoverable structural information—even with missing nodes—and that a carefully constructed LS2-based formulation can extract this structure under reasonable conditions. The behaviour across diffusion, full SI, and partial SI also highlights where the method is well-posed, where it becomes initialization-dependent, and where richer latent modelling may be required in future work.

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